



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.NOS.8

Compare and Order Integers

Lesson 9-3 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can compare and order integers using a number line.

Language Objective: I can explain my reasoning using the words compare, order, greater than, and less than.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Number line

Compare

Order

Greater than

Less than

Negative integer

Integer



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

To plot integers in order, place them on a .

2

To decide which number is larger or smaller is to them.

3

To arrange numbers from least to greatest or greatest to least is to them.

4

On a number line, a number farther to the right is another number.

5

On a number line, a number farther to the left is another number.

6 An integer less than zero, like -6 , is a .

7 A whole number or its opposite is an .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Order these temperatures from coldest to warmest: -8, 3, -1, 5. How does the number line help you put them in order?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Number line** — A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.
Recta numérica — Una recta donde los números se colocan en orden; los números son menores a la izquierda y mayores a la derecha.

☐ **Compare** — To see if a number is bigger, smaller, or equal to another.
Comparar — Ver si un número es mayor, menor o igual que otro.

☐ **Order** — To put numbers in order, smallest to biggest or biggest to smallest.
Ordenar — Poner números en orden, de menor a mayor o de mayor a menor.

☐ **Integer** — Whole numbers and their opposites, like -2, -1, 0, 1, 2.
Número entero — Números enteros y sus opuestos, como -2, -1, 0, 1, 2.

☐ **Greater than** — Farther right on a number line. The > sign.
Mayor que — Más a la derecha en la recta numérica. El signo >.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The coldest temperature is ____ because it is farthest ____.

La temperatura mas fría es ____ porque esta mas a la ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: -3 is less than 2 because on the number line ____.

Evidence: Student explains that -3 is farther left on the number line than 2, and numbers decrease moving left.

Reasoning: This evidence connects the definition of compare to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The coldest temperature is ____ because it is farthest ____.**

La temperatura mas fria es ____ porque esta mas a la ____.

- **From least to greatest, the order is ____.**

De menor a mayor, el orden es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Compare
3. **Notes 3:** Order
4. **Notes 4:** Greater than
5. **Notes 5:** Less than
6. **Notes 6:** Negative integer
7. **Notes 7:** Integer
8. **Watch for this mistake:** *A common mistake in Compare and Order Integers is thinking that a negative number with a bigger digit is the bigger value — for example, believing -15 is greater than -8 because 15 is greater than 8. It's the opposite for negatives: the number farther to the left on the number line (farther from zero) is always the LESSER value, so $-15 < -8$. Before you submit, check your comparison against the number line, not just the digits.*
9. **Try It 1:** A. -4 m — *Closer to zero means greater for negatives. -4 is only 4 below the surface, so -4 is the greatest of -9, -4, and -15.*
10. **Try It 2:** A. $<$ — *-8 is farther left (deeper) than -3, so -8 is less than -3. The symbol is $<$.*